

Frontier Model Performance Across 20+ Benchmarks

A comparative analysis of 100+ models across agentic coding, frontier reasoning, safety alignment, and multimodal capabilities.

20+

benchmarks

100+

models

4

capability domains

Agentic

coding + tool use

Frontier

reasoning + knowledge

Safety

alignment + honesty

Multimodal

audio + vision-language

20+ Benchmarks Spanning Four Capability Domains Evaluate 100+ Models

Scale's suite combines private datasets for overfitting resistance with open-source datasets for comparability.

Scale SEAL

OpenAI

Google

Anthropic

Muse Spark

Agentic

Coding + tool use

- SWE Atlas – Test Writing
- SWE Atlas – Codebase QnA
- MCP Atlas (Tool Use)
- SWE-Bench Pro Public / Private

OpenAI leads most coding; Muse Spark wins tool use.

Frontier

Reasoning + knowledge

- Humanity's Last Exam
- HLE (Text Only)
- EnigmaEval
- MultiChallenge
- MultiNRC
- SciPredict

Top scores remain far below saturation.

Safety

Alignment + honesty

- MASK (Honesty)
- PropensityBench
- Fortress
- Professional Reasoning Legal / Finance
- Remote Labor Index

Claude dominates safety + professional reasoning.

Multimodal

Audio + vision-language

- AudioMultiChallenge Text
- AudioMultiChallenge Audio
- VISTA
- VisualToolBench

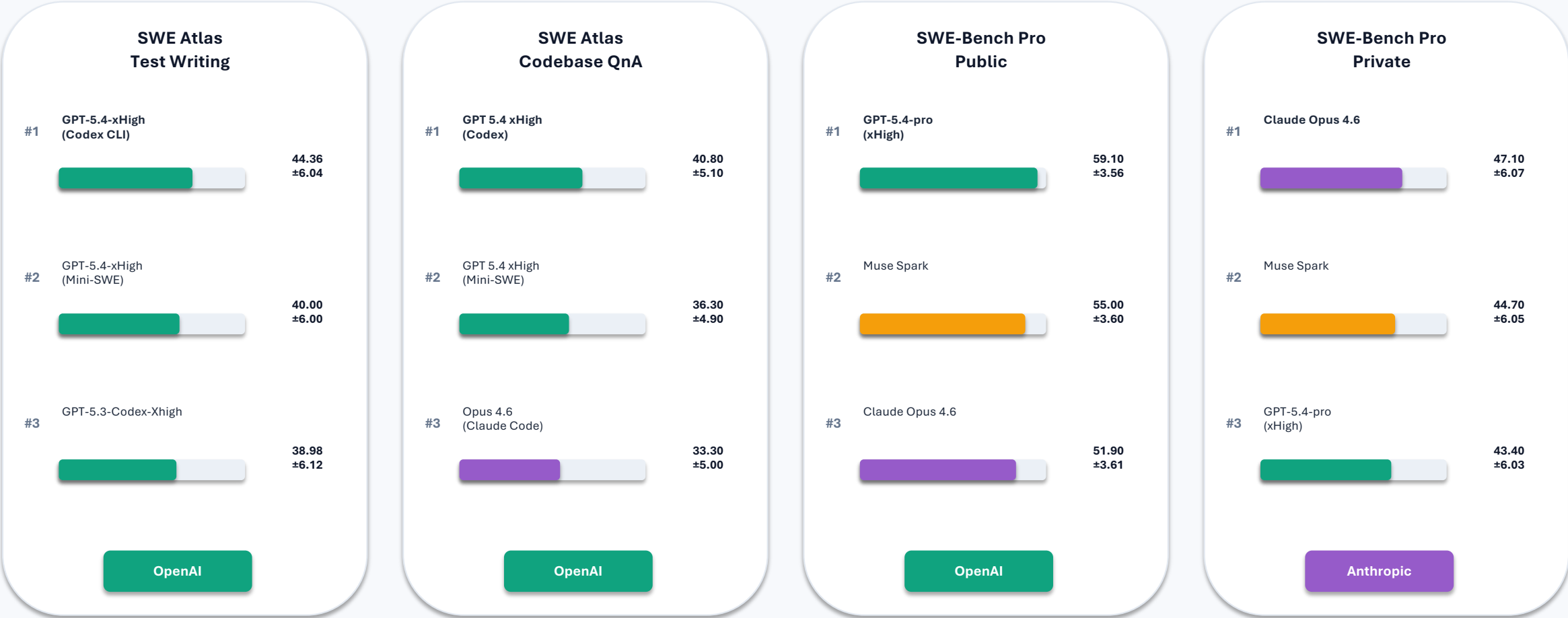
Gemini sweeps audio and leads VISTA.

GPT-5.4 Dominates Agentic Coding: Top Scores on SWE Atlas and SWE-Bench Pro

Top 3 comparison across SWE Atlas and SWE-Bench Pro subtasks; scores are percentages with confidence intervals.

OpenAI Google Anthropic Muse Spark

Scale SEAL



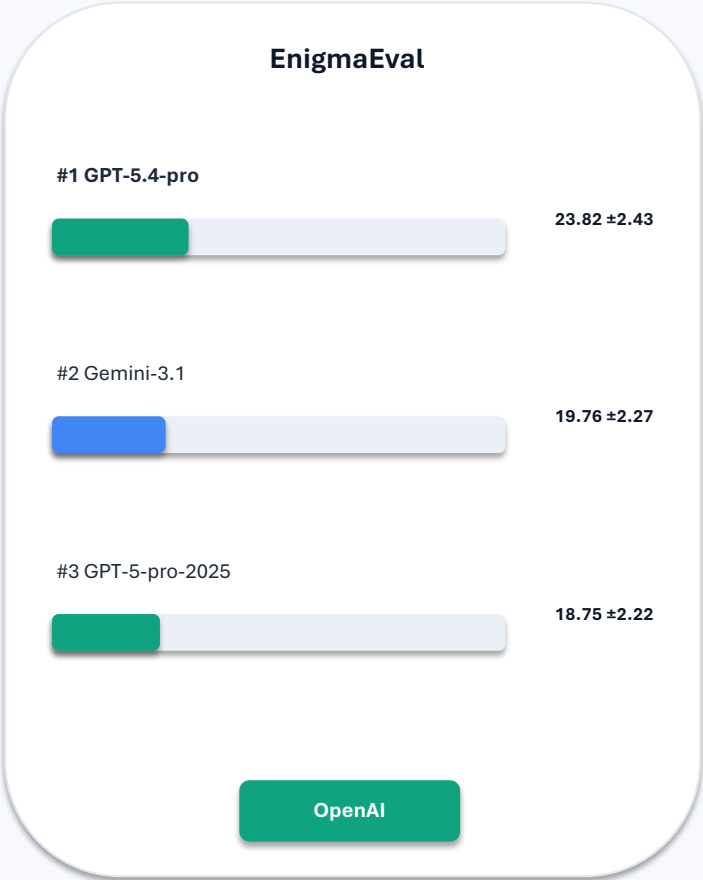
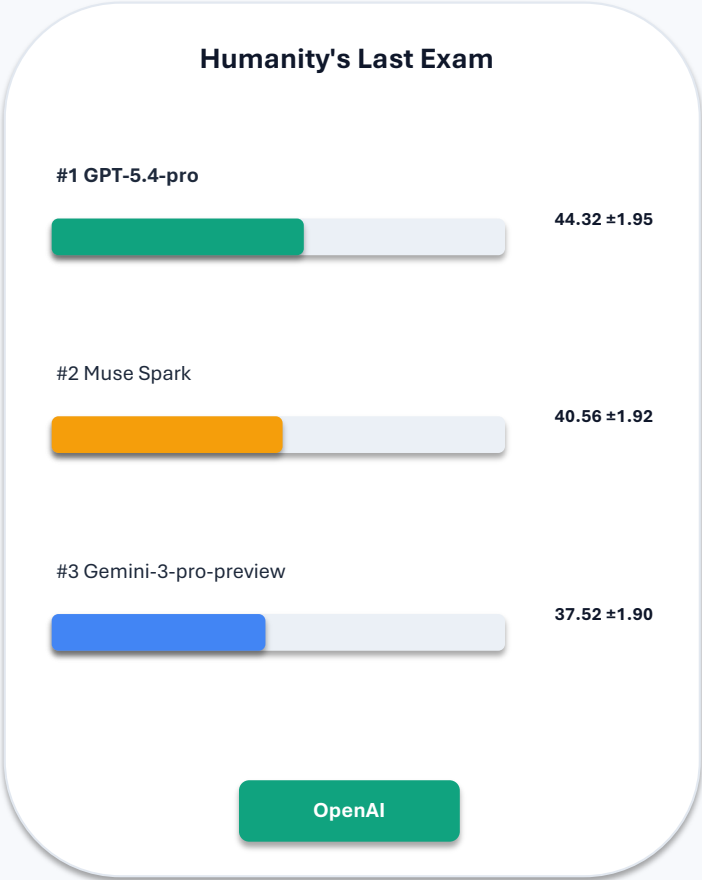
Pattern: OpenAI GPT-5.4 variants take #1 on three of four required coding subtasks; Claude Opus 4.6 leads private repositories, with Muse Spark consistently in contention on SWE-Bench Pro.

GPT-5.4-Pro Leads Humanity's Last Exam at 44.3% — Scores Remain Below 50%

Reasoning leaderboards show persistent headroom: even #1 models are far from saturation on HLE and EnigmaEval.

● OpenAI ● Google ● Anthropic ● Muse Spark

Scale SEAL



Unsaturated frontier: HLE tops out at 44.32%, EnigmaEval at 23.82%, and SciPredict at 25.27% — complex reasoning and forecasting remain open problems.

Gemini Models Sweep Audio and Vision-Language Benchmarks

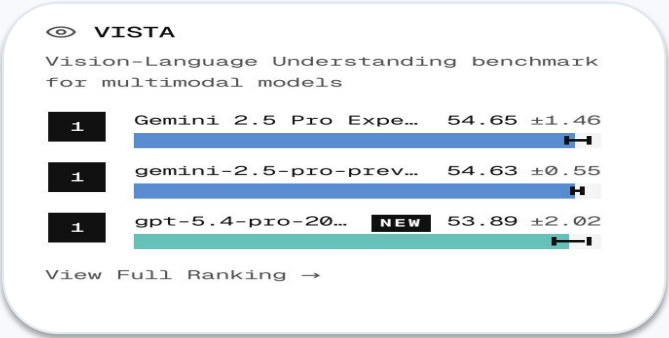
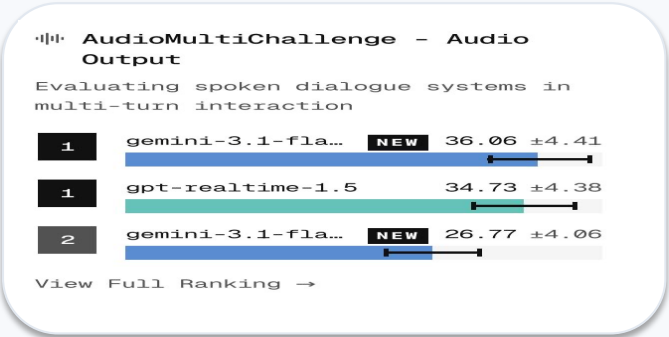
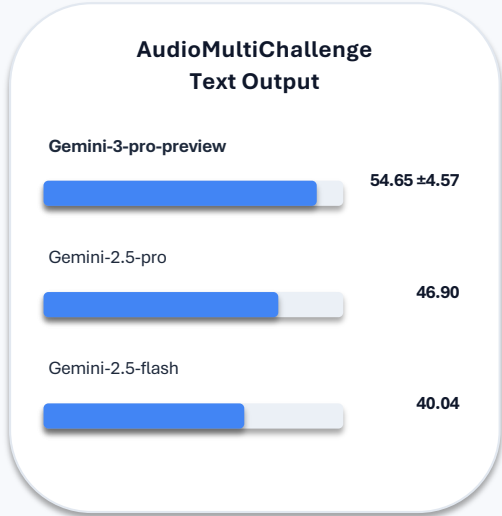
Scale SEAL

Google’s Gemini variants hold the lead on AudioMultiChallenge text/audio output and VISTA; VisualToolBench is a close OpenAI edge.

Google

Anthropic

Muse Spark



VisualToolBench: GPT-5.4 edges Gemini-3.1-pro, 29.17 vs 28.97.

Takeaway

Gemini’s sweep of audio and VISTA suggests a multimodal-specialized advantage, while OpenAI remains highly competitive on tool-mediated visual reasoning.

Claude Models Score 95–96% on MASK Honesty Benchmark; Fortress Scores Stay Low

Alignment performance is highly task-dependent. Near-ceiling honesty does not imply high scores on every safety benchmark.

Scale SEAL

OpenAI Google Anthropic Muse Spark

Near-ceiling honesty: MASK



Anthropic holds the top two positions on MASK, both above 96%.

Low-score safety tasks: Fortress + PropensityBench



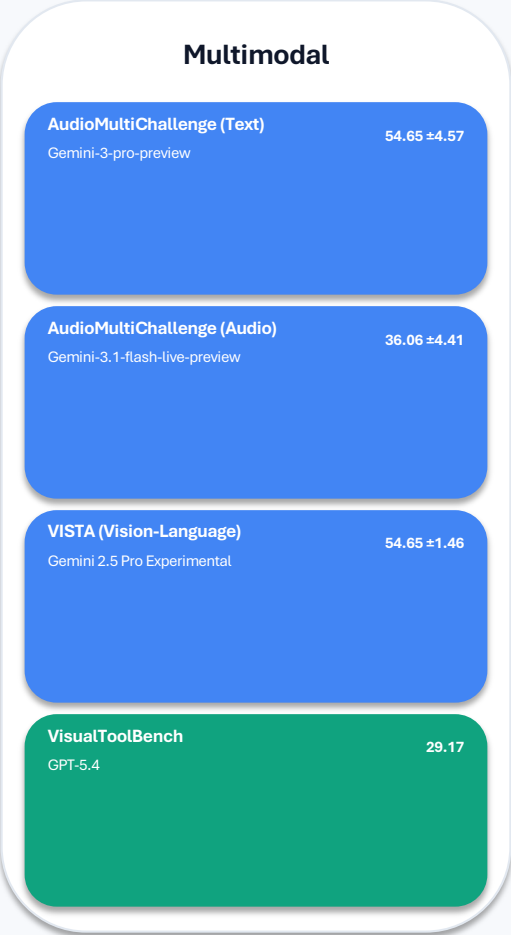
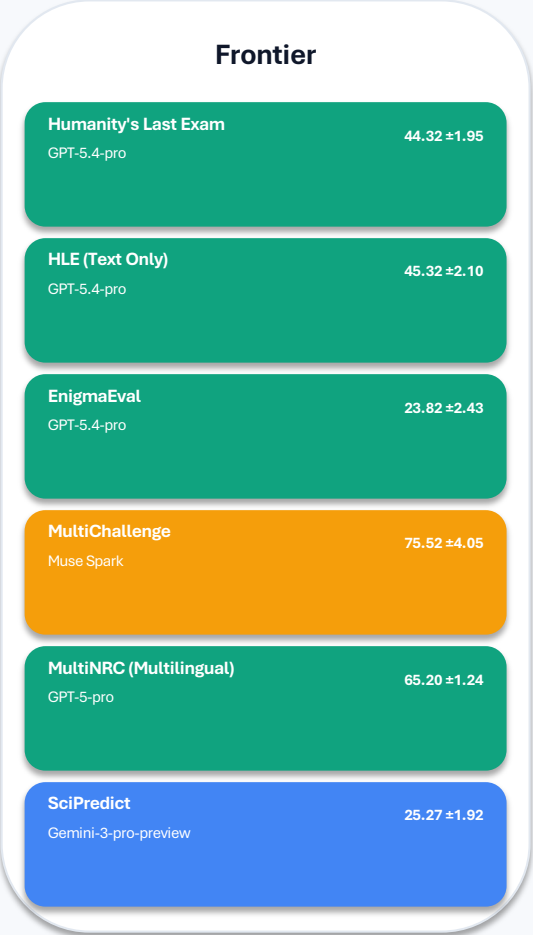
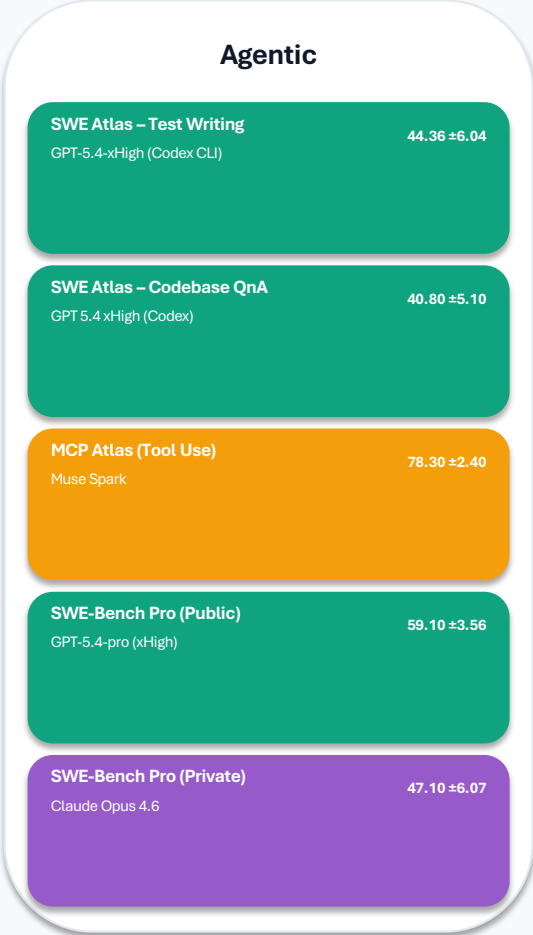
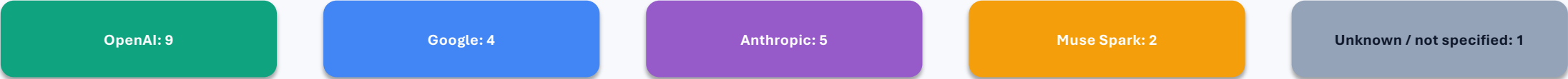
Variance: 96%+ on MASK vs 8–13 range on Fortress; “alignment” is not a single scalar capability.

OpenAI, Google, and Anthropic Each Claim #1 Across Different Domains

Scale SEAL

Heatmap-style winner matrix across every benchmark named in the brief; cell color indicates the #1-ranked lab/model.

● OpenAI ● Google ● Anthropic ● Muse Spark



Key Takeaways

No single model wins everything — specialization is the new frontier.

GPT-5.4 leads agentic/reasoning — not multimodal

OpenAI wins SWE Atlas, SWE-Bench Pro Public, HLE, EnigmaEval, MultiNRC, and VisualToolBench.

Gemini dominates audio/vision

Gemini variants lead AudioMultiChallenge Text, AudioMultiChallenge Audio, and VISTA.

Claude leads safety + professional reasoning

Claude Opus 4.6 reaches 96.28% on MASK and ranks #1 on Legal, Finance, and RLI.

Hard benchmarks remain unsaturated

HLE tops at 44.32%, EnigmaEval at 23.82%, and SciPredict at 25.27%.

Muse Spark is a serious new entrant

Top-3 on 6+ benchmarks and #1 on MultiChallenge at 75.52%.

Implication for model evaluation

- Portfolio view beats single-score ranking
- Benchmark choice should match deployment domain
- Private + open evaluations remain complementary
- New entrants can win targeted capability niches

Conclusion: frontier capability is fragmenting by modality, task type, and alignment objective.